

LLMs on Tabular Data

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Introdução

- Dados tabulares amplamente utilizados (Fang et al., 2024)
- Desafio: adaptar LLMs ao formato tabular (Sui et al., 2024).
- Abordagens: integrar raciocínio e entendimento semântica aos dados estruturados (Dong, 2024).

Características dos Dados Tabulares

- Estrutura: linhas e colunas, dados heterogêneos (numéricos, categóricos, textuais)
- Esparsidade e valores ausentes
- Interdependência entre atributos
- Ordem invariante (não sequencial)
- Falta de conhecimento estrutural prévio
- Tipos: relacionais, hierárquicos, layout/visuais

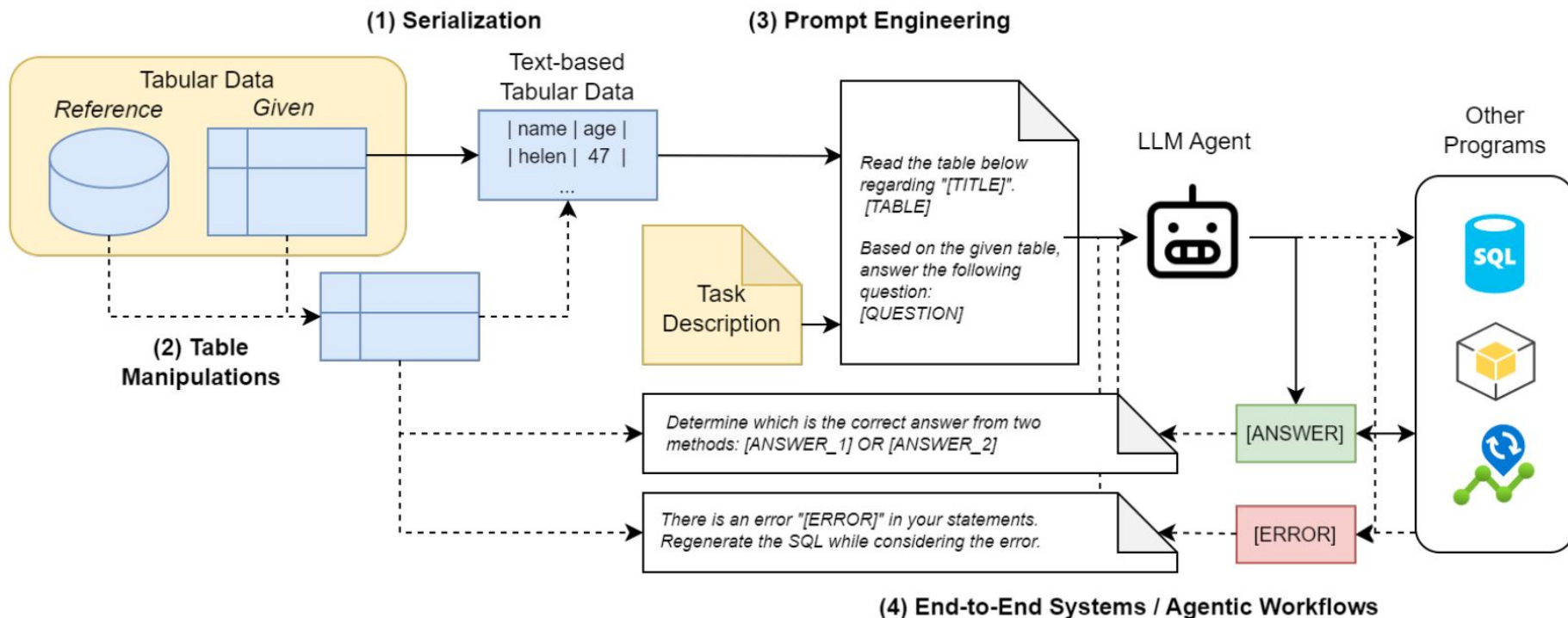
Características de Modelos de ML

Tabular input characteristics	Prediction	Data Generation	Table Understanding
	<u>Classical</u>		
- Heterogeneity - Sparsity - Lack of locality - Dependency on preprocessing - Contextual interconnection - Lack of prior knowledge	- Tree-based ensemble - GBDT - e.g. XGBoost, LightGBM, and CatBoost	- Copulas - Bayesian networks	- Rule based - Hand-crafted features
	<u>Deep learning</u>		
	- Data transformation - e.g. SuperTML, IGTD, 1D-CNN - e.g. Wide&Deep, DeepFM, DNN2LR - Differentiable trees - e.g. NODE, SDTR, Net-DNF, DeepGBM, TabNN, BGNN - Attention-based - e.g. TabNet, TabTransformer, TransTab, ARM-net, SAINT, NPT - Regularization - e.g. RLN, Regularization Cocktails	- VAEs - GANs - Diffusion	- e.g. Seq2SQL, SQLizer, SQLNet, TYPESQL, TAPAS, TableQuery

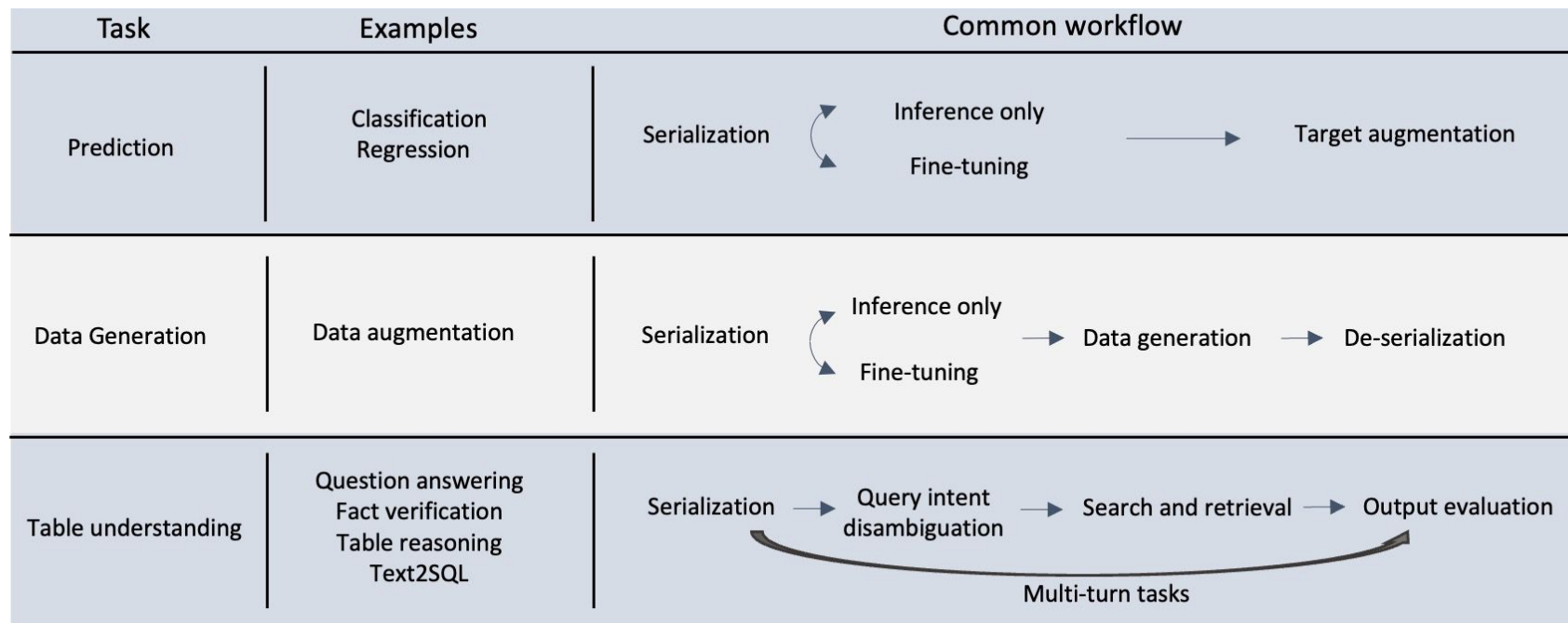
Técnicas Tradicionais

- Modelos baseados em árvores: GBDT, XGBoost, LightGBM, CatBoost
- Redes neurais: Wide&Deep, DeepFM, TabTransformer, TabNet, SAINT
- Modelos generativos: Copulas, VAEs, GANs
- Limitações: engenharia de features, generalização, dados esparsos

Técnicas chave das LLMs nos dados tabulares



LLMs em Dados Tabulares



Aplicações usuais das LLMs

- Compreensão e geração de texto
- Aprendizado em contexto (ICL), Chain-of-Thought
- Instrução-tuning e multitarefas
- Agentes autônomos e ferramentas externas

LLMs com Dados Tabulares

- Serialização: JSON, Markdown, HTML, CSV
- HTML/XML: melhor desempenho por exposição prévia
- Prompt engineering e few-shot examples
- Fine-tuning / Instruction-tuning: QA, Text2SQL, Data Generation
- Modelos: TABERT, TAPAS, TableGPT, TableLlama, TableLLM

Estado da Arte

- Benchmarks: SUC Benchmark (Sui et al., 2024), TableLLM, TableGPT3
- Desempenho: GPT-4 > GPT-3.5, mas ainda limitado
- Abordagens:
 - Self-augmentation
 - Retrieval-Augmented Generation (RAG)
 - Multimodalidade (texto + imagem)

SUC Benchmark

SUC Benchmark	Stages	Capabilities	Tasks
	Partition & Parsing	Structural Description Detection	Table Partition
		Format Understanding	Table Size Detection
			Hierarchy Detection
	Search & Retrieval	Grounding/Locating	Cell Lookup & Reverse Lookup
		Operation Reasoning	Column & Row Retrieval

Resultados do teste de ablation

Format	TabFact	HybridQA	SQA	Feverous	ToTTo
	Acc	Acc	Acc	Acc	BLEU-4
NL + Sep	70.26%	45.02%	70.41%	75.15%	12.70%
Markdown	68.40%	45.88%	66.59%	71.88%	8.57%
JSON	68.04%	42.40%	70.39%	73.84%	8.82%
XML	70.00%	47.20%	70.74%	73.14%	8.82%
HTML	71.33%	47.29%	71.31%	75.20%	12.30%
GPT-4 w/ HTML	78.40%	56.68%	75.35%	83.21%	20.12%

Desafios

- Limite de contexto e custo computacional
- Alucinação e inconsistência estrutural
- Serialização sensível ao formato
- Falta de benchmarks padronizados
- Raciocínio quantitativo deficiente
- Escalabilidade e interpretabilidade

Trabalhos Futuros

- Modelos híbridos (LLM + GBDT)
- Representações numéricas aprimoradas
- Benchmarks padronizados e datasets abertos
- LLM-agents com raciocínio iterativo
- Multimodalidade (texto, imagem, tabela)
- Self-verification e auto-raciocínio

Conclusão

- LLMs ampliam o potencial sobre dados estruturados
- Persistem limitações em precisão e compreensão estrutural
- Tendência: integração multimodal, interpretabilidade, padronização

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Referências

DONG, H.; WANG, Z. Large language models for tabular data: Progresses and future directions. In: Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval. New York, NY, USA: Association for Computing Machinery, 2024. (SIGIR '24), p. 2997-3000. ISBN 9798400704314. Disponível em: <https://doi.org/10.1145/3626772.3661384>.

FANG, X. et al. Large language models (llms) on tabular data: Prediction, generation, and understanding - a survey. Transactions on Machine Learning Research, 2024. ISSN 2835-8856. Disponível em: <https://openreview.net/forum?id=IZnrCGF9WI>.

SUI, Y. et al. Table meets llm: Can large language models understand structured table data? a benchmark and empirical study. In: Proceedings of the 17th ACM International Conference on Web Search and Data Mining. New York, NY, USA: Association for Computing Machinery, 2024. (WSDM '24), p. 645-654. ISBN 9798400703713. Disponível em: <https://doi.org/10.1145/3616855.3635752>.

Table meets LLM: Can large language models understand structured table data? A benchmark and empirical study (Sui, et. al. 2024)

- LLMs are becoming attractive for few-shot reasoning in NL-related tasks.
- A benchmark called SUC evaluates LLMs' structural understanding capabilities.
- Performance varies depending on input format, content order, and prompting.
- Self-augmentation for effective structural prompting leads to promising improvements.
- The SUC benchmark and proposed methods can serve as future research selections.

LLMs on tabular data: Prediction, generation, and understanding - a survey (Fang, *et al.* 2024)

- LLMs are being explored for tabular data tasks like prediction and generation.
- Tabular data presents challenges due to heterogeneity, sparsity, and pre-processing needs.
- Serialization methods convert tabular data into text for LLM input.
- Prompt engineering techniques like in-context learning and Chain-of-Thought enhance LLM performance.
- LLMs are applied in medical, financial, and recommendation prediction, and data generation.

Large Language Models for tabular data: Progresses and future directions (Dong, 2024)

- The tutorial provides a comprehensive study of advances, challenges, and opportunities in leveraging LLMs for tabular data.
- It introduces methods for prompting or training LLMs for table interpreting and processing.
- The tutorial covers table representation, training LLMs, tasks, and benchmarks for tabular data.
- It also explores LLM-driven table agents and future research directions.
- The content is highly relevant to table data retrieval, machine learning, and data intelligence.

Antigos

Características (Fang, *et al.* 2024)

1. Heterogeneidade
2. Esparsidade
3. Dependência de pré-processamento
4. Interconexão baseada em contexto
5. Ordem invariante
6. Ausência de conhecimento prévio

O problema

- Geração de dados
- Anonimização
- Ordem de linhas irrelevantes
- Dados colunares relevantes
- Números VS Tokenização

Soluções

- Serialização (Fang, *et al.* 2024)
 - CSV, TSV, JSON, MD, HTML, XML
 - Texto sequencial
 - Text Based
 - Embedding Based
 - Graph-based & Tree Based
 - Comparisons
- Tokens de estrutura (Sui, *et. al.* 2024)
 - <HEAD>, <ROW>
- 0-Shot VS 1-Shot (Sui, *et. al.* 2024)

Estado da arte 🎨 (Fang, *et al.* 2024)

- Gradient-boosted decision trees
- Boosting ensemble
- Recurrent Neural Networks (RNNs)
- Deep Learning
 - a. Data Transformation
 - b. Differentiable Trees
 - c. Attention-based methods
 - d. Regularization Methods
- Data Synthesis
- Table understanding

LLMs (Fang, *et al.* 2024)

- Statistical Language Models (SLM)
- Neural Language Models (NLM)
- Pretrained Language Models (PLM)
- Large Language Models (LLMs)
 - In-context Learning
 - Instruction following
 - Multi-step reasoning

LLMs in tabular data (Fang, *et al.* 2024)

- BERT e variações
- Contribuições (Fang, *et al.* 2024)
 - A formal break down of key techniques for LLMs' applications on tabular data
 - A survey and taxonomy of metrics for LLMs' applications on tabular data
 - A survey and taxonomy of datasets for LLMs' applications on tabular data
 - A survey and taxonomy of techniques for LLMs' applications on tabular data
 - An overview of future research directions

Feitos por GPT abaixo